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Foreword

This European Telecommunication Standard (ETS) has been produced by the Special Mobile Group (SMG) Technical Committee (TC) of the European Telecommunications Standards Institute (ETSI).

This ETS defines the stage 2 of the Advice of Charge (AoC) supplementary services for the European digital cellular telecommunications system (Phase 2).

This ETS corresponds to GSM Technical Specification (GSM-TS) GSM 03.86 version 4.6.1.

The specification from which this ETS has been derived was originally based on CEPT documentation, hence the presentation of this ETS may not be entirely in accordance with the ETSI/PNE rules.

Reference is made within this ETS to GSM-TSs (NOTE).

NOTE: TC-SMG has produced documents which give the technical specifications for the implementation of the European digital cellular telecommunications system. Historically, these documents have been identified as GSM Technical Specifications (GSM-TSs). These TSs may have subsequently become I-ETSs (Phase 1), or ETSs (Phase 2), whilst others may become ETSI Technical Reports (ETRs). GSM-TSs are, for editorial reasons, still referred to in current GSM ETSs.

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0 General

0.1 Scope

This specifications gives the stage 2 description of the advice of charge supplementary services.

The charging supplementary services currently defined are:

- Advice of Charge (Information) (AoCI) (section 1);
- Advice of Charge (Charging) (AoCC) (section 2).

0.2 Normative references

This ETS incorporates by dated and undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references, the latest edition of the publication referred to applies.

[1] GSM 01.04 (ETR 100): "European digital cellular telecommunications system (Phase 2); "Abbreviations and acronyms".

[2] GSM 02.24 (ETS 300 510): "European digital cellular telecommunications system (Phase 2); Description of Charge Advice Information (CAI)".

[3] GSM 02.86 (ETS 300 519): "European digital cellular telecommunications system (Phase 2); Advice of Charge (AoC) supplementary services - Stage 1".

[4] GSM 03.11 (ETS 300 529): "European digital cellular telecommunications system (Phase 2); Technical realization of supplementary services".

0.3 Definitions and abbreviations

Abbreviations used in this specification are listed in GSM 01.04.

1 Advice of Charge (Information) (AoCI)

1.1 Advice of Charge (Information) MSC SDL diagram

The SDL diagram for the Advice of Charge (Information) supplementary service within the MSC is given in figure 1.1.

NOTE 1: AoC (Information) is not applicable to emergency calls.

NOTE 2: The request of generation of the Charge Advice Information (CAI) applies to AoC initiation or charge modification.

1.2 Advice of Charge (Information) mobile equipment SDL diagram

The SDL diagram for the Advice of Charge (Information) supplementary service within the mobile equipment is given in figure 1.2. This SDL indicates when charging calculations are started, amended and stopped, within the mobile equipment. The mobile equipment should start charging calculations as soon as possible after receiving the charging information. Charging calculations should be amended by the mobile equipment in accordance with TS GSM 02.24.

NOTE: The point at which the charging calculations are amended can occur before or after sending the AoC Acknowledge.

Charging calculations are stopped when the call ends for any reason.

The reception of the CAI shall be acknowledged only if the MS supports the AoCI functionality specified in TS GSM 02.24 and GSM 02.86, however the network does not action this acknowledgment in the AoC (Information) case.

1.3 Advice of Charge (Information) information flow diagram

The Advice of Charge (Information) information flow diagrams are shown in figure 1.3.

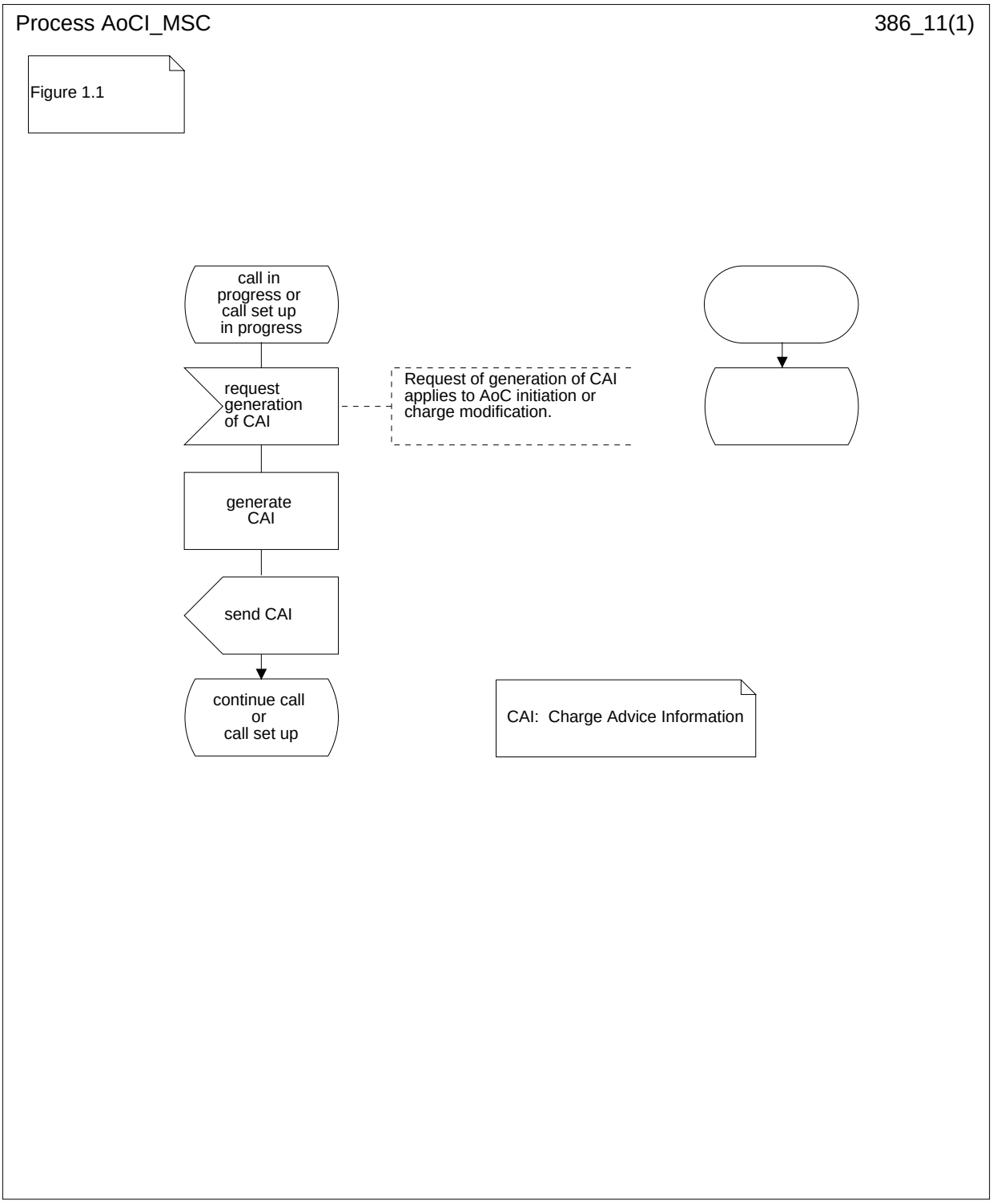


Figure 1.1: SDL diagram of advice of charge (information) in the MSC

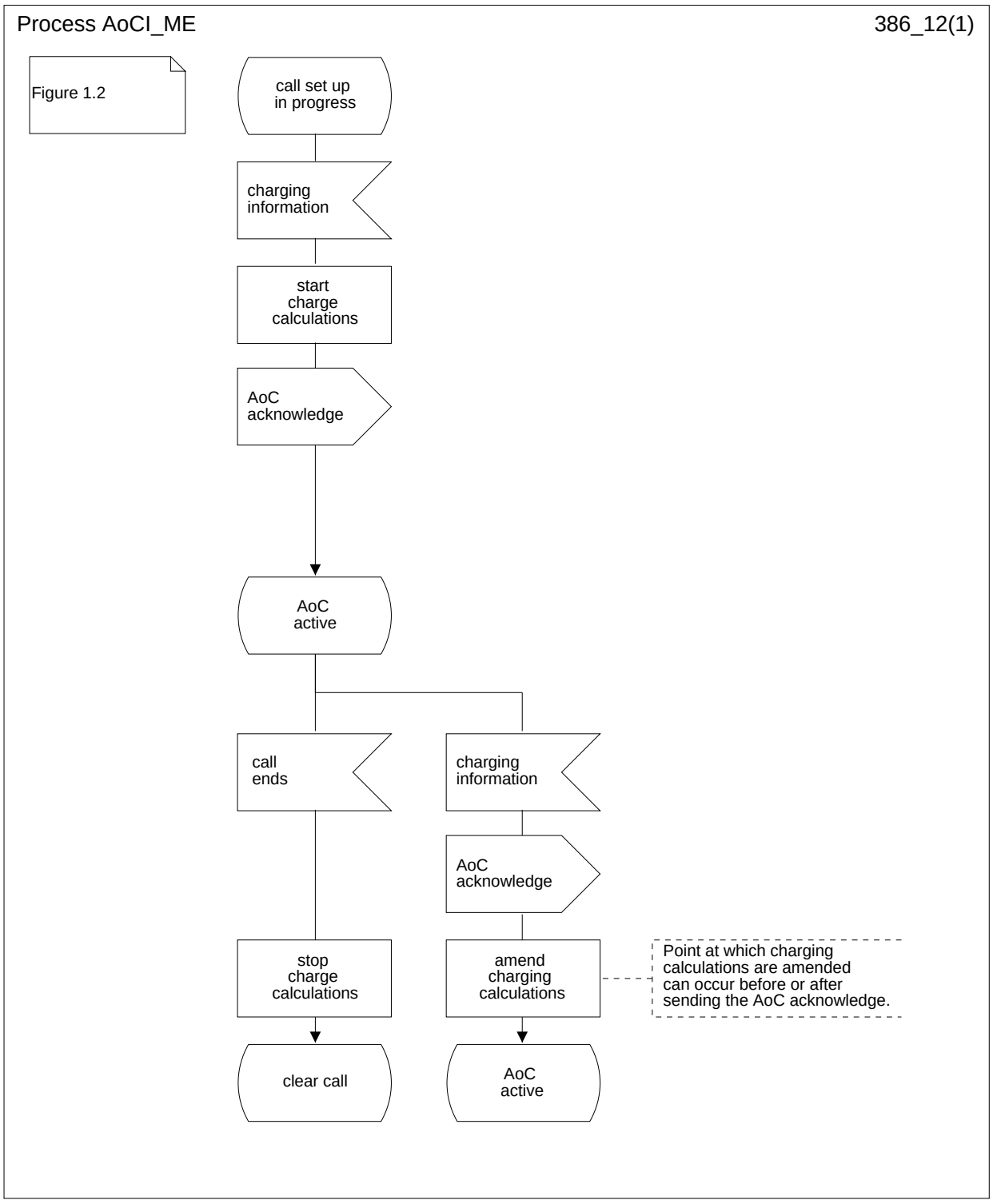


Figure 1.2: SDL diagram of advice of charge (information) in the mobile equipment

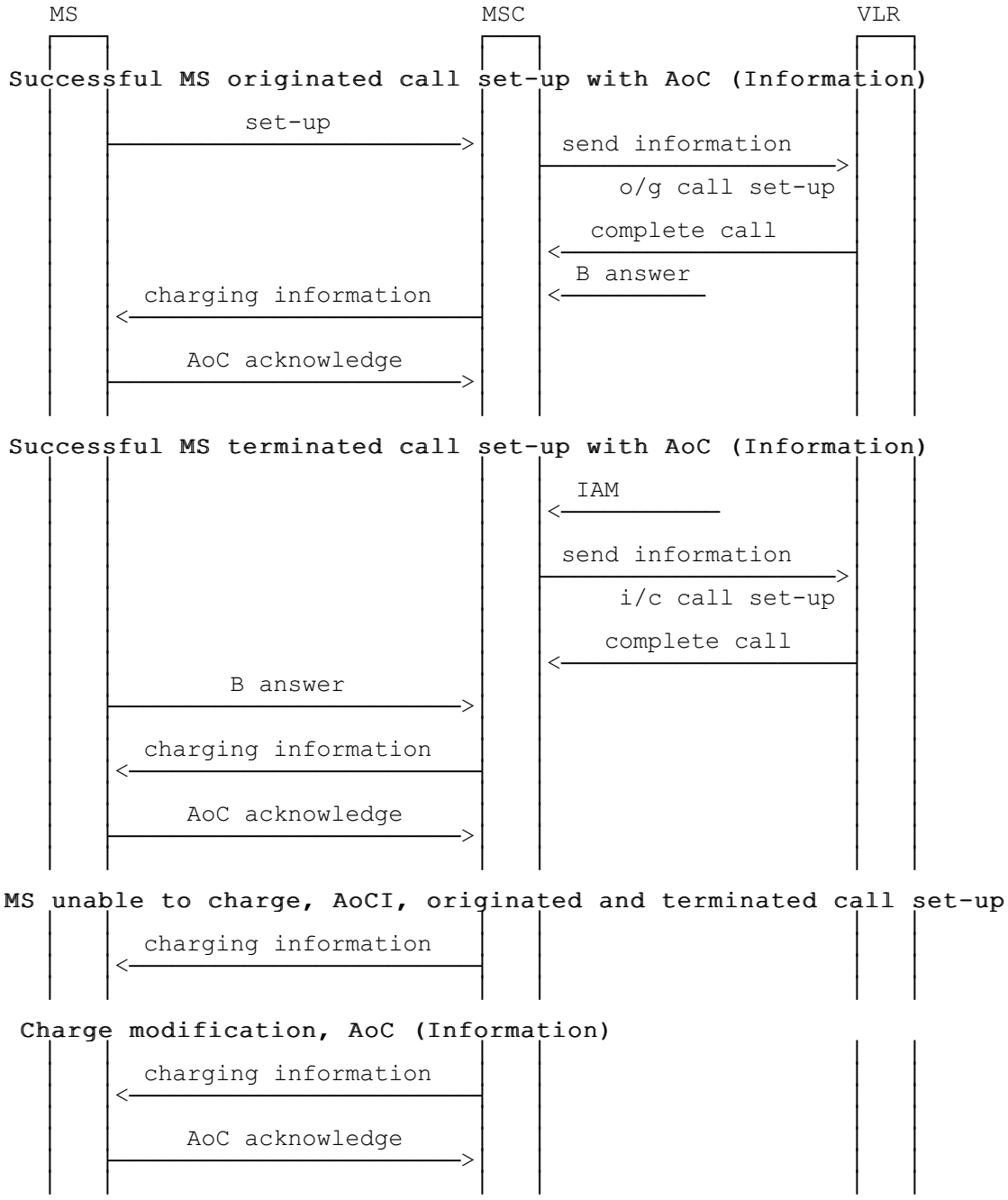


Figure 1.3: Information flow for Advice of Charge (Information)

1.4 Information stored in the HLR

AoCI may have the following logical states (refer to TS GSM 03.11 for an explanation of the notation):

Provisioning State	Registration State	Activation State	HLR Induction State
(Not Provisioned,	Not Applicable,	Not Active,	Not Induced)
(Provisioned,	Not Applicable,	Active and Operative,	Not Induced)

The HLR shall store the logical state of AoCI (which shall be one of the valid states listed above) on a per subscriber basis.

1.5 State transition model

The following figure shows the successful cases of transition between the applicable logic states of AoCI. The state changes are caused by actions of the service provider.

Note that error cases are not shown in the diagram as they normally do not cause a state change. Additionally, some successful requests may not cause a state change. Hence they are not shown in the diagram.

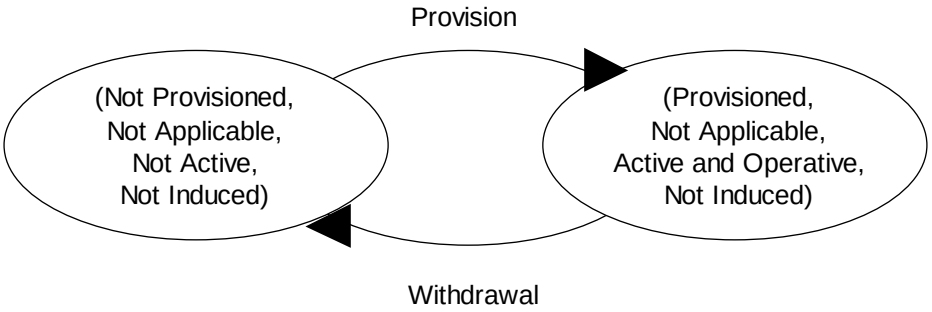


Figure 1.4: State transition model for AoCI

1.6 Transfer of Information from HLR to VLR

If the provisioning state for AoCI is "Provisioned" then when the served subscriber registers on a VLR the HLR shall send that VLR information about the logical state of AoCI.

If the logical state of AoCI is changed while a subscriber is registered on a VLR, then the HLR shall inform the VLR of the new logical state of AoCI.

1.7 Information stored in the VLR

For the supplementary service AoCI the VLR shall store the service state information received from the HLR.

1.8 Handover

Handover will have no impact on AoC control procedure.

2 Advice of Charge (Charging) (AoCC)

2.1 Advice of Charge (Charging) MSC SDL diagram

The SDL diagram for the Advice of Charge (Charging) supplementary service within the MSC is given in figure 2.1. At invocation of the Advice of Charge (Charging) supplementary service the network shall send the "Charging" MS the CAI. The network shall allow the call to proceed whilst waiting for an acknowledgement of the CAI. The waiting period is governed by Timer T(AoC). On expiry of T(AoC) the network shall initiate call clearing.

NOTE 1: AoC (Charging) is not applicable to emergency calls.

NOTE 2: The request of generation of CAI applies to AoC initiation or charge modification.

2.2 Advice of Charge (Charging) mobile equipment SDL diagram

The SDL diagram for the Advice of Charge (Charging) supplementary service within the mobile equipment is given in figure 2.2. The SDL indicates when charging calculations are started, amended and stopped, within the mobile equipment. The mobile equipment should start charging calculations as soon as possible after receiving the charging information. Charging calculations should be emended by the mobile equipment in accordance with TS GSM 02.24.

NOTE: The point at which charging calculations are amended can occur before or after sending the AoC Acknowledge.

Charging calculations are stopped when the call ends for any reason.

The reception of the CAI shall be acknowledged only if the MS supports the AoCC functionality specified in TS GSM 02.24 and GSM 02.86.

For mobile originated calls, the MS shall prevent a call set-up attempt if the ACM value is equal to or greater than ACMmax.

For mobile terminated calls, a call set-up attempt shall be allowed even if the ACM value is equal to or greater than ACMmax. In this case, when the MS detects that this is a chargeable call and that the ACM value is equal to or greater than ACMmax then the call shall be released.

During an active call, when the MS detects that ACM is equal to or greater than ACMmax then the MS shall release the call.

2.3 Advice of Charge (Charging) information flow diagram

The Advice of Charge (Charging) information flow diagrams are shown in figure 2.3.

2.4 AoCC subscriber roaming requirements

When an AoCC subscriber first roams into an MSC area not supporting AoCC, or the subscription is given to him while he is roaming in an MSC area not supporting AoCC the following applies:

The HLR shall indicate to the VLR that roaming is restricted in this MSC area due to unsupported feature, and it prevents further mobile terminated traffic. In this case the MS is not reachable.

When entering an AoCC supporting area the AoCC related roaming restriction shall be removed in the HLR.

When AoCC is withdrawn the AoCC related roaming restriction shall be removed in the HLR and VLR. This is independent from roaming restrictions due to other reasons.

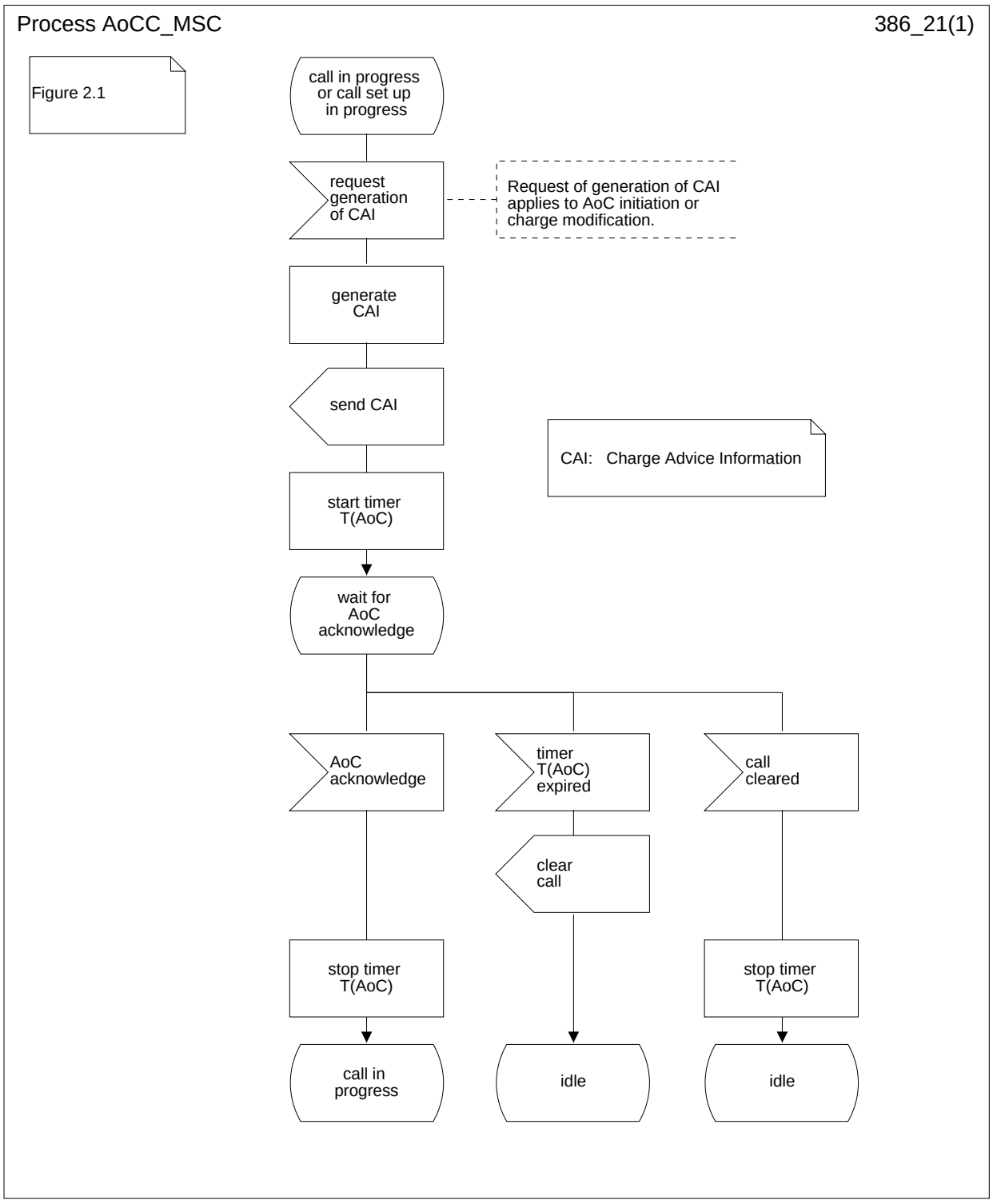


Figure 2.1: SDL diagram of advice of charge (charging) in the MSC

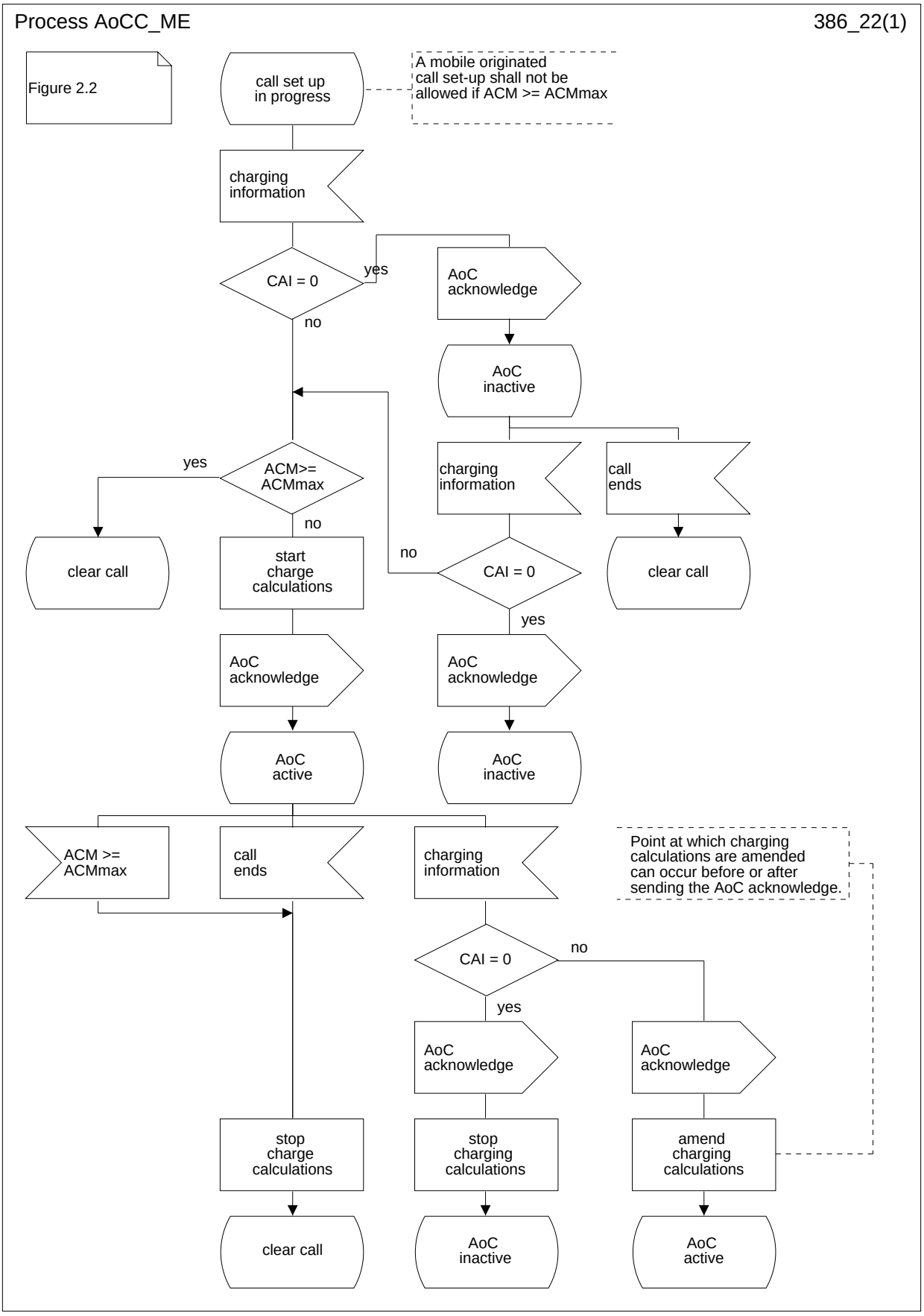


Figure 2.2: SDL diagram of advice of charge (charging) in the mobile equipment

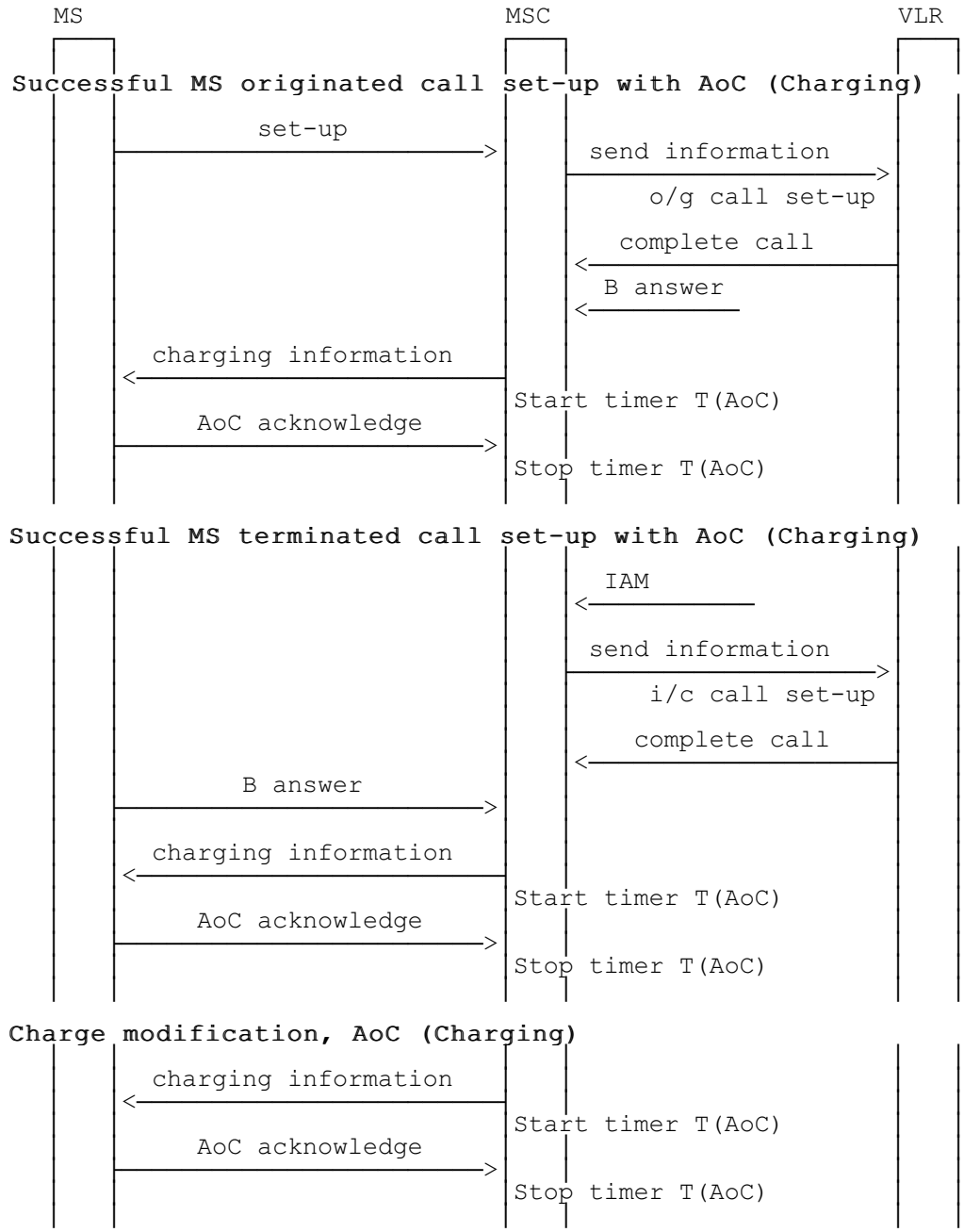


Figure 2.3 (sheet 1 of 2): Information flow for Advice of Charge (Charging)

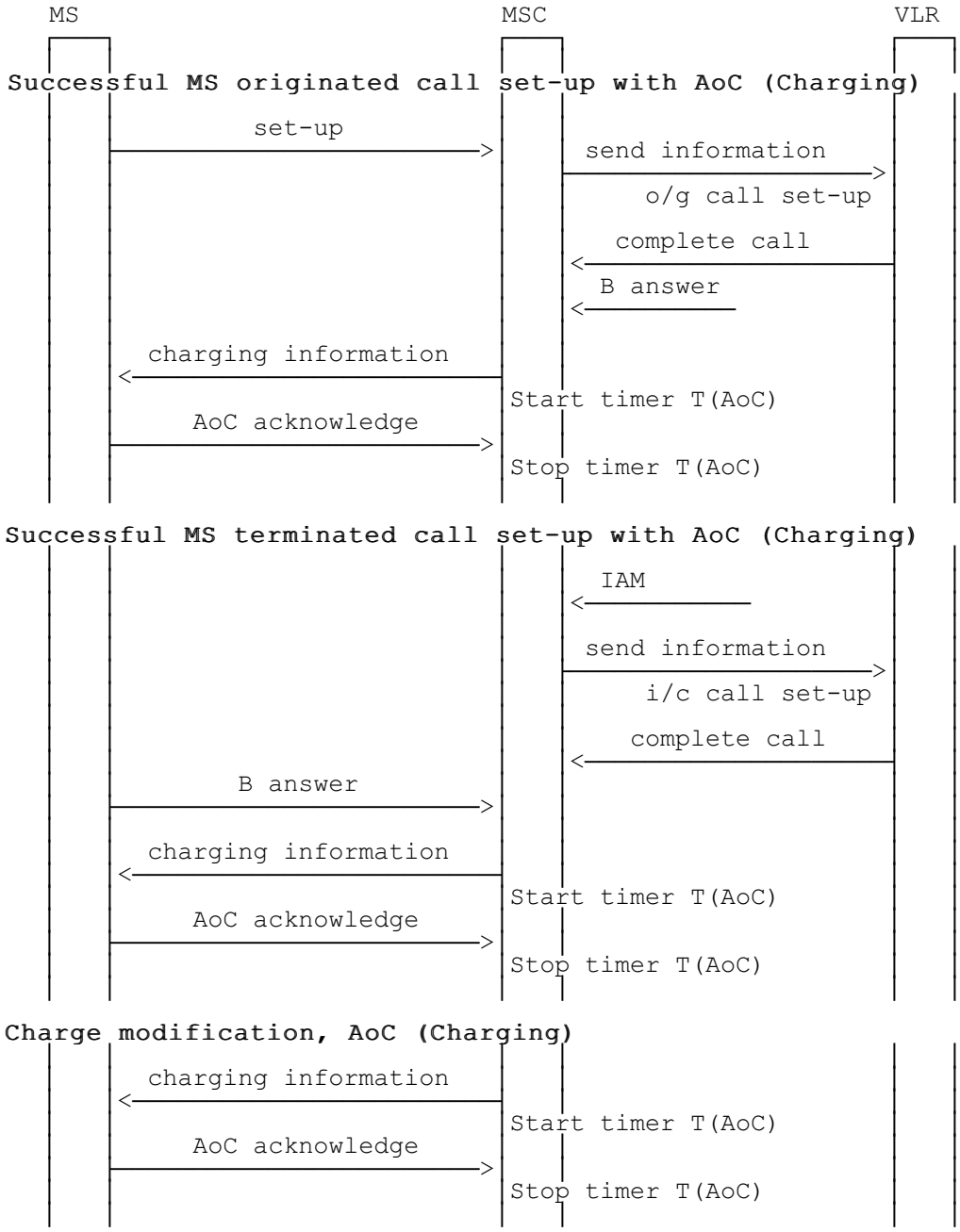


Figure 2.3 (sheet 2 of 2): Information flow for Advice of Charge (Charging)

2.5 Information stored in the HLR

AoCC may have the following logical states (refer to TS GSM 03.11 for an explanation of the notation):

Provisioning State	Registration State	Activation State	HLR Induction State
(Not Provisioned,	Not Applicable,	Not Active,	Not Induced)
(Provisioned,	Not Applicable,	Active and Operative,	Not Induced)

The HLR shall store the logical state of AoCC (which shall be one of the valid states listed above) on a per subscriber basis.

2.6 State transition model

The following figure shows the successful cases of transition between the applicable logic states of AoCC. The state changes are caused by actions of the service provider.

Note that error cases are not shown in the diagram as they normally do not cause a state change. Additionally, some successful requests may not cause a state change. Hence they are not shown in the diagram.

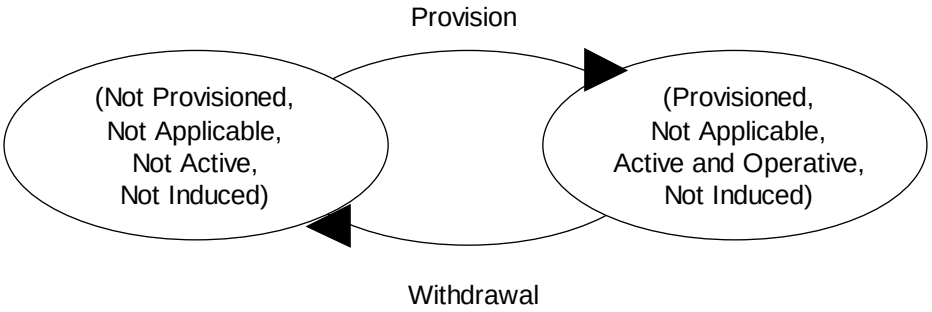


Figure 2.4: State transition model for AoCC

2.7 Transfer of Information from HLR to VLR

If the provisioning state for AoCC is "Provisioned" then when the served subscriber registers on a VLR the HLR shall send that VLR information about the logical state of AoCC.

If the logical state of AoCC is changed while a subscriber is registered on a VLR, then the HLR shall inform the VLR of the new logical state of AoCC.

2.8 Information stored in the VLR

For the supplementary service AoCC the VLR shall store the service state information received from the HLR.

2.9 Handover

Handover will have no impact on AoC control procedure.

Annex A (normative): Information stored in the MSC

The MSC shall have access to the following tables to enable the call handling function to generate the Charge Advice Information (CAI), see TS GSM 02.24. The table will give the CAI element values corresponding to service request, call destination, type of day and the time of day.

- Units per interval table (e1 values);
- Seconds per internal table (e2 values);
- Scaling factor table (e3 values);
- Unit increment table (e4 values);
- Units per data interval table (e5 values);
- Segments per data interval table (e6 values);
- Initial seconds per time interval table (e7 values).

History

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